Detekcija vozila

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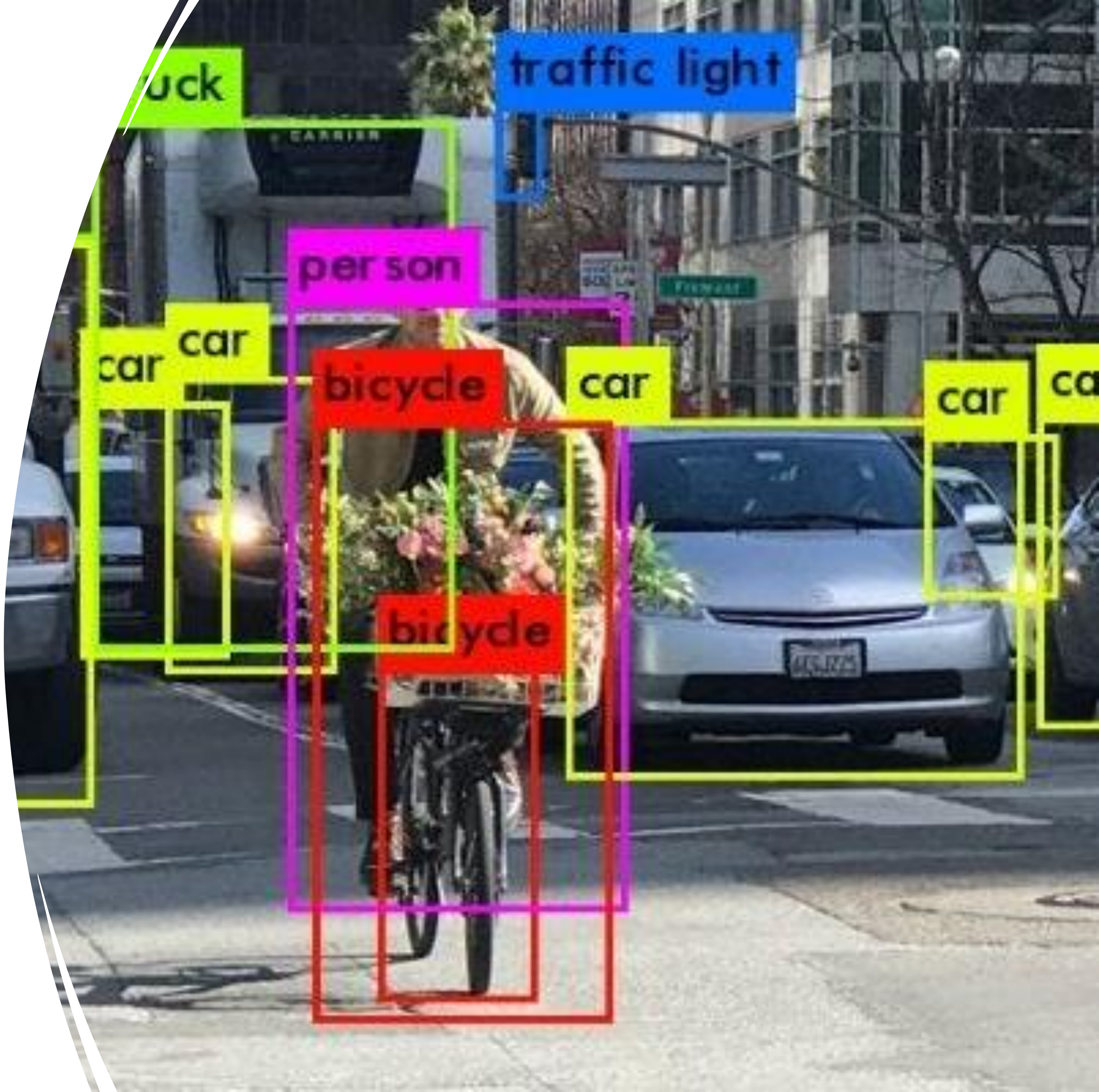
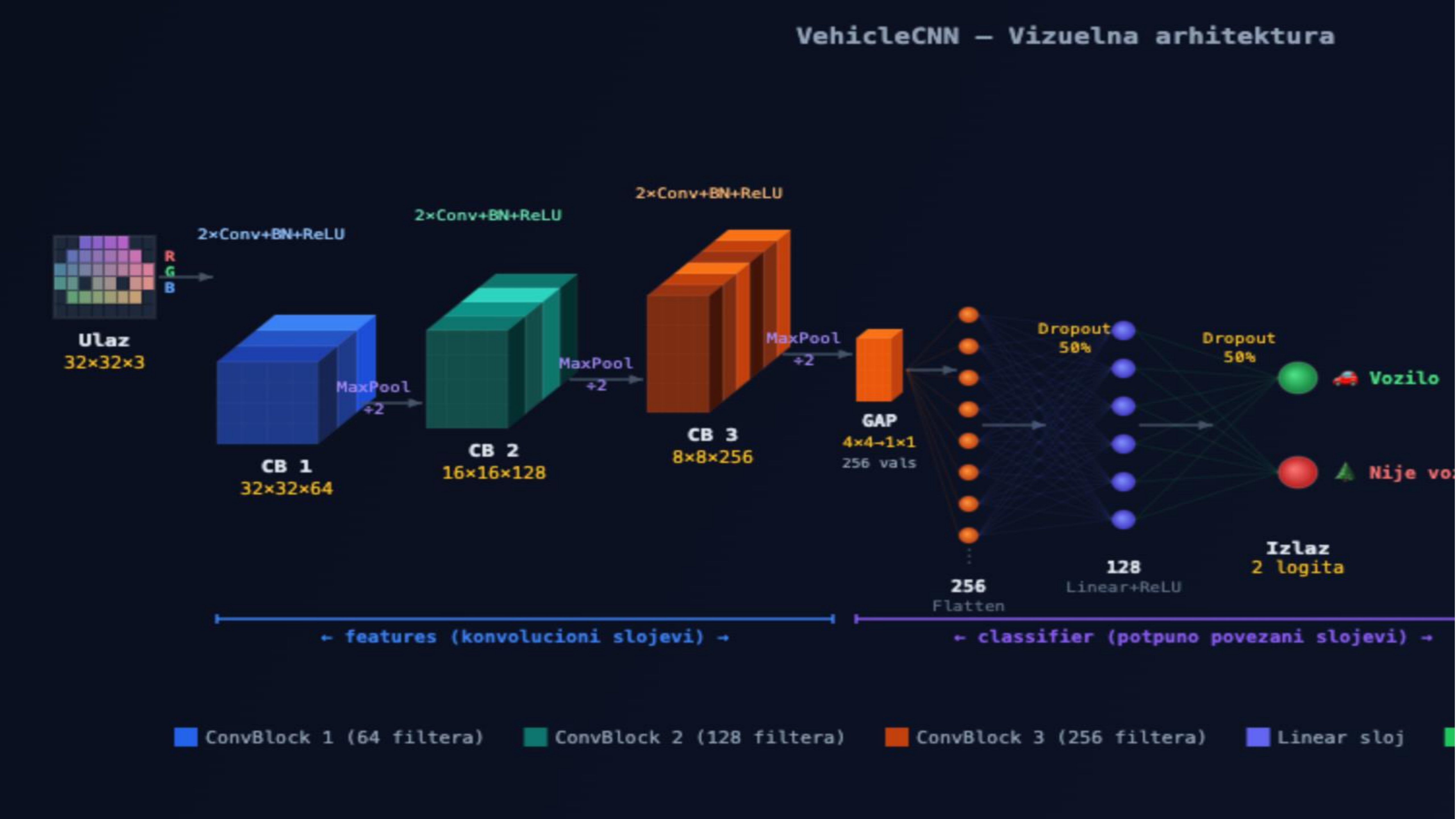
Uvod

- Cilj projekta: odrediti da li slika sadrži vozilo.
- Primene su beskrajne, od kamera i senzora do dronova i autonomnih vozila
- Akutelna tema



Skup podataka i pretprocesiranje

- **CIFAR-10**: 60.000 slika, 32×32 RGB, 10 klasa (50k trening / 10k test)
- **Binarno mapiranje**: vozilo = avion, automobil, brod, kamion (4 klase) vs nije vozilo = ptica, mačka, jelen, pas, žaba, konj (6 klasa)
- **Podela train skupa**: 90% trening / 10% validacija
- **Crop & Flip**
- **Normalizacija**
- **Augmentacija** trening skupa


VehicleCNN — Vizuelna arhitektura
R
G
B
Ulaz
32×32×3
2×Conv+BN+ReLU
CB 1
32×32×64
MaxPool
÷2
2×Conv+BN+ReLU
CB 2
16×16×128
MaxPool
÷2
2×Conv+BN+ReLU
CB 3
8×8×256
MaxPool
÷2
GAP
4×4→1×1
256 vals
256
Flatten
Dropout
50%
128
Linear+ReLU
Dropout
50%
Vozilo
Nije
Izlaz
2 logita
← features (konvolucioni slojevi) →
← classifier (potpuno povezani slojevi) →
ConvBlock 1 (64 filtera)
ConvBlock 2 (128 filtera)
ConvBlock 3 (256 filtera)
Linear sloj

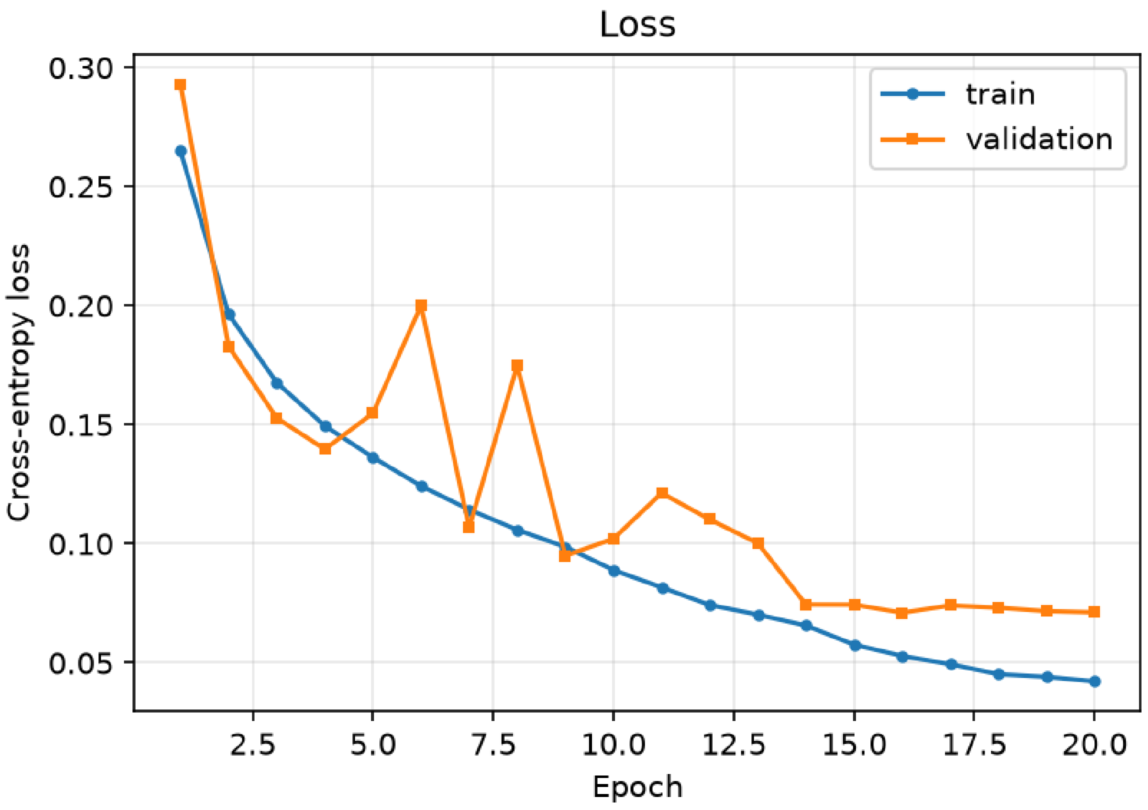
Treniranje

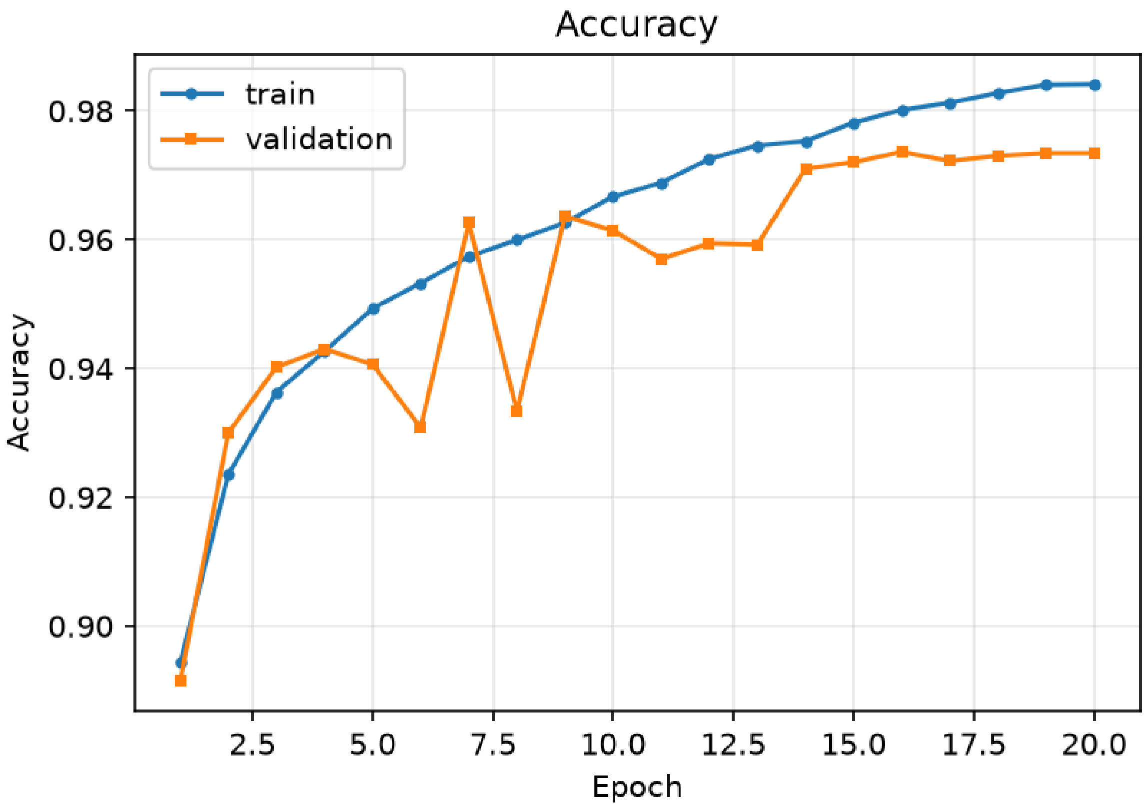
- **Optimizer i gubitak**
- Optimizer: AdamW (lr = 0.001, weight_decay = 1e-4)
- Loss funkcija: CrossEntropyLoss
- Scheduler: CosineAnnealingLR (T_max = 20 epoha)
- **Hiperparametri**
- Batch size: 128 | Epohe: 20 | Seed: 42
- Augmentacija: RandomCrop(32, padding=4) + RandomHorizontalFlip

Treniranje

- Validacioni skup: 10% trening podataka (5 000 slika)
- **Čuvanje modela**
- Checkpoint se čuva svaki put kad val_acc pobije dosadašnji maksimum
- Finalna evaluacija: best model učitan na test skupu (10 000 slika)

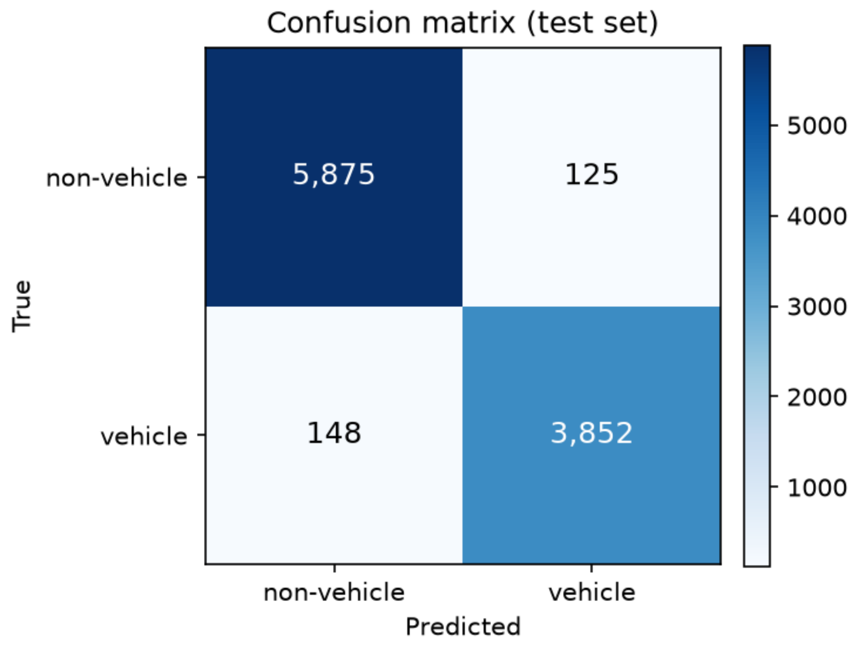
Krive učenja



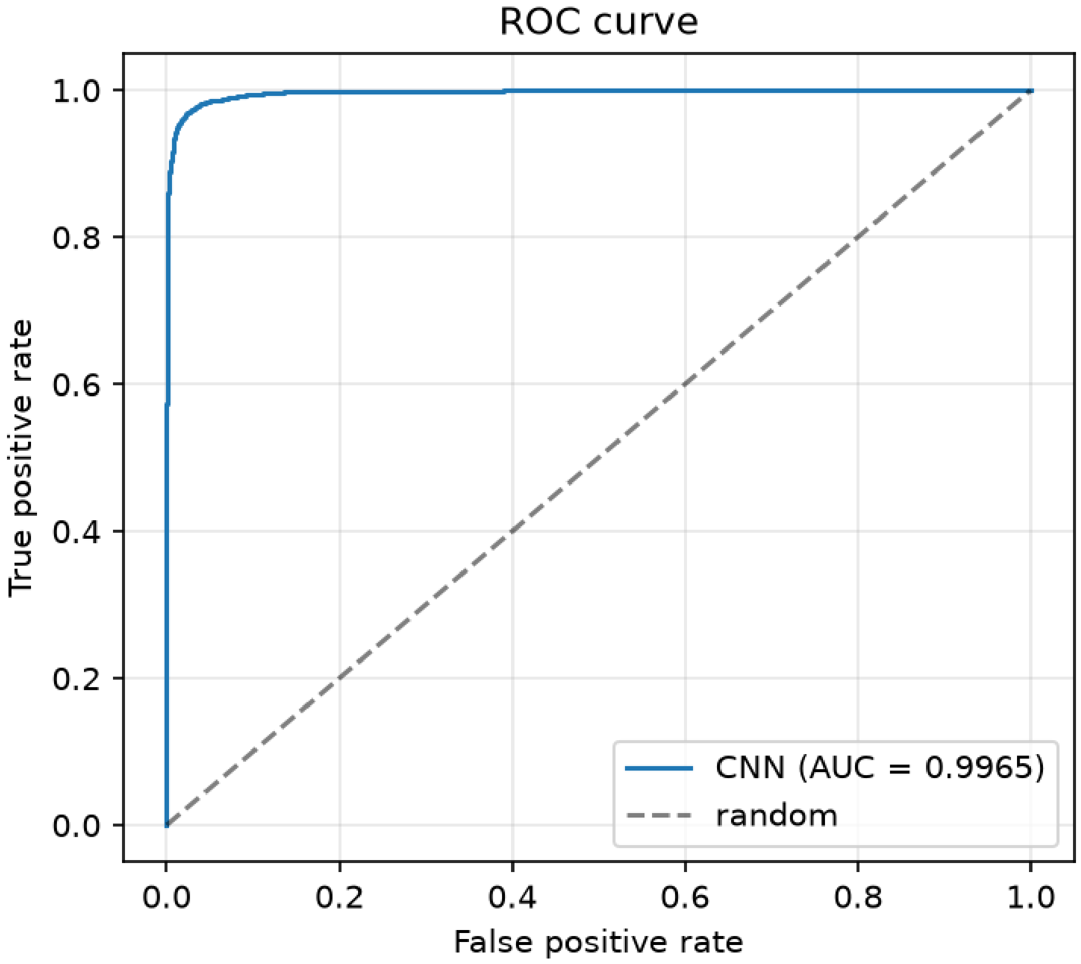


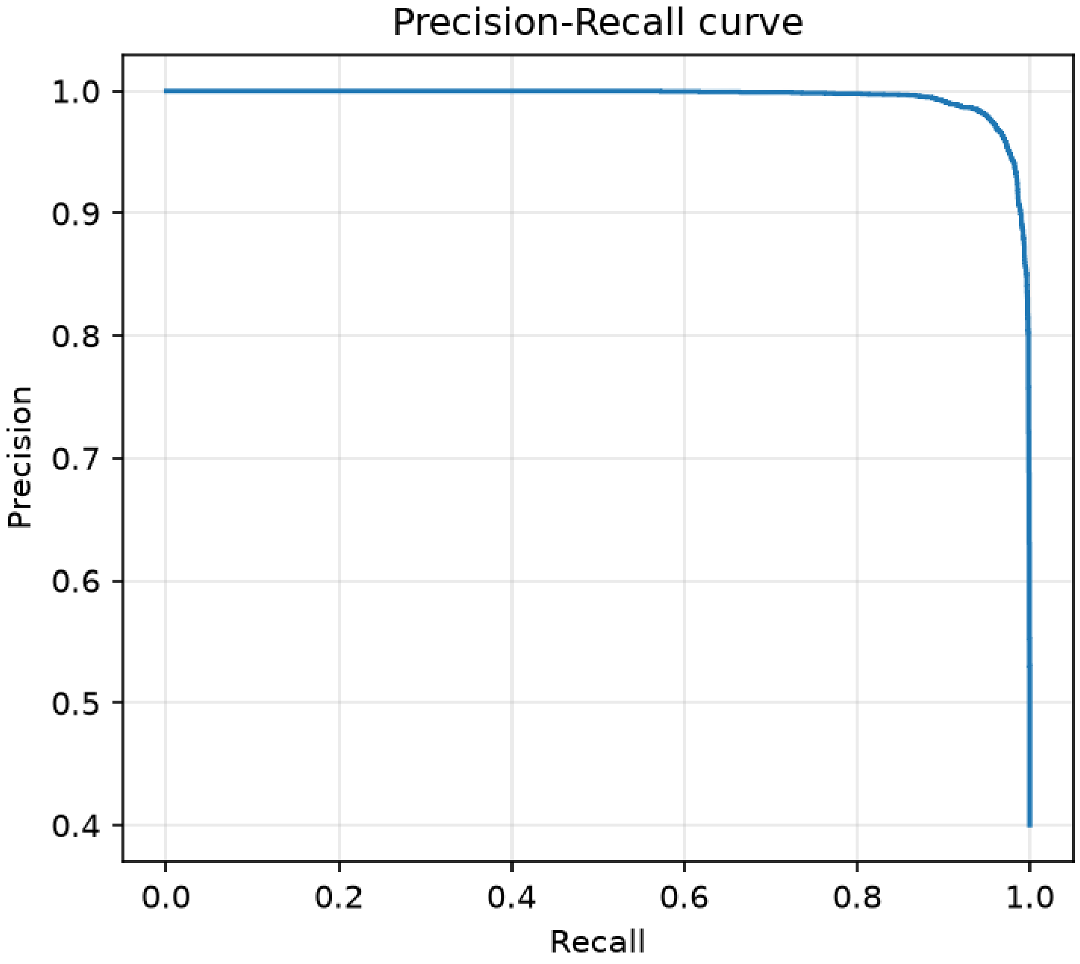
Evaluacija na test skupu

97.27%	**96.86%**	**96.30%**	**96.58%**	**99.65%**
Tačnost	Preciznost	Odziv	F1 skor	ROC-AUC

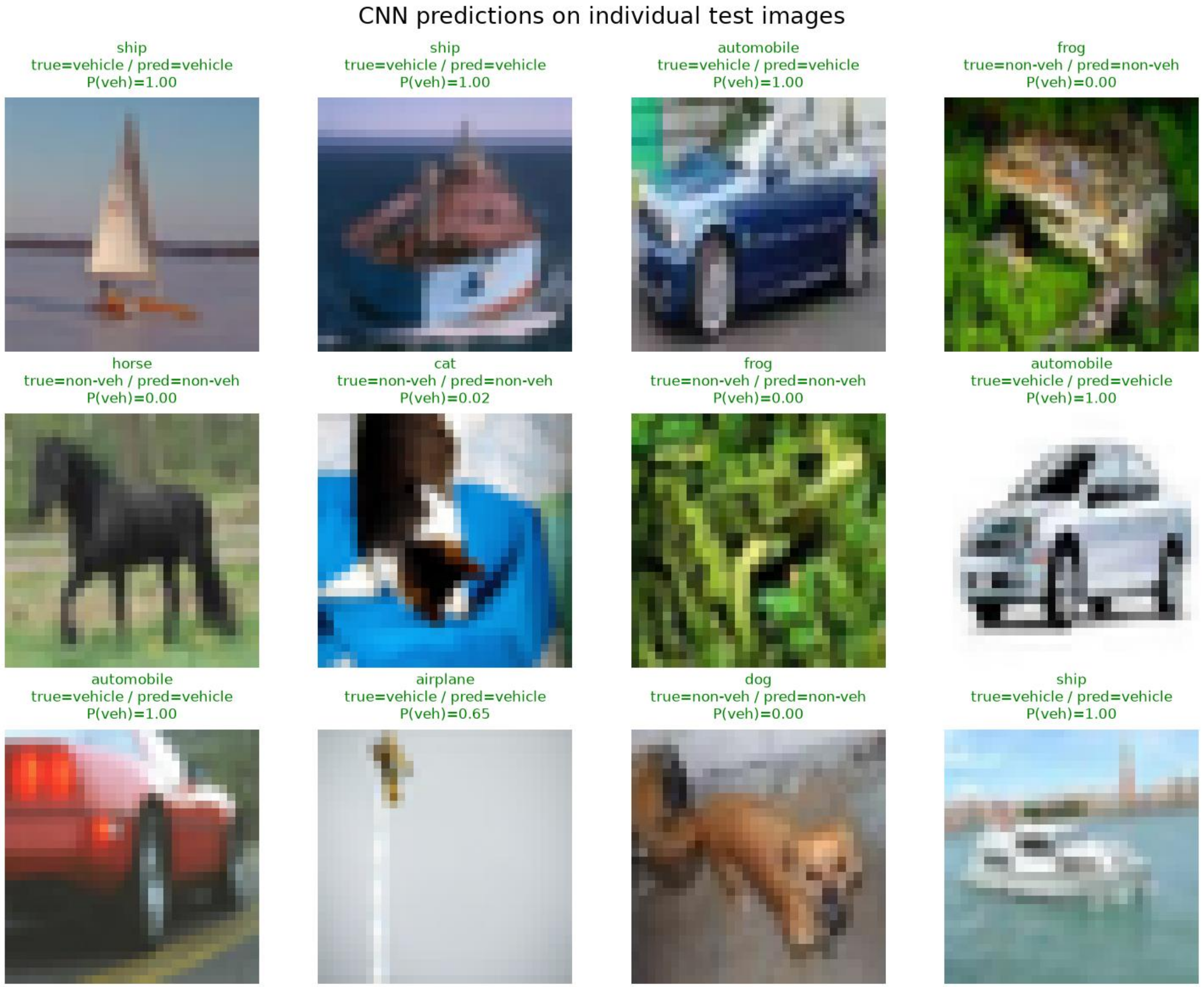


Analiza rezultata





Primeri predikcija modela



Poređenje sa baseline modelima

Model	Tačnost	F1	ROC-AUC	Params
VehicleCNN	97.27%	96.58%	99.65%	1 180 354
MLP (bez conv)	88.04%	84.97%	94.34%	—
Logistička regresija	81.56%	76.08%	87.90%	—